RISHIKESH R

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EXECUTIVE SUMMARY

Security-focused CSE student with strengths in **offensive testing, system internals, and low-level analysis**, building **detection systems and secure backend pipelines**.

TECHNICAL SKILLS

Kernel & Systems: C/C++, Linux Kernel Modules (LKM), kprobes, Netlink, Syscall Internals, Socket Programming
Security: VAPT, OWASP Top 10, Network Pentesting, Digital Forensics, Log Analysis & Network Traffic Analysis
Tools: Burp Suite, Metasploit, Nmap, Wireshark, SQLmap, Volatility
Core CS: Operating Systems, Computer Networks, Data Structures & Algorithms, OOP, Cryptography
Backend & Data: Python, FastAPI, Flask, REST APIs, PostgreSQL, MongoDB
Platforms & Infrastructure: Linux (Ubuntu, Kali), Docker, Git/GitHub, VMware, Windows

EXPERIENCE

Alcatel-Lucent Enterprise (ALE India Pvt. Ltd.) [GitHub] Bangalore, India
Software Development Intern July 2025 – Aug 2025

- Built a **FastAPI-based log analysis platform** to ingest, parse, and visualize AOS switch and HMON datasets, enabling TSEs to perform efficient log analysis and anomaly investigation across **10K+ log entries** by module, timestamp, and case.
- Developed **log ingestion pipelines** to automate extraction from nested TAR archives, parse multi-file switch logs using regex into structured JSON, and generate **Plotly-based dashboards** for CPU, memory, and fan telemetry.

PROJECTS

SENTINEL-X — Cross-Platform Kernel Integrity Monitor Sep 2025 – Present
Kernel Engineering • Distributed Systems • Security

- Built a **cross-platform kernel-level rootkit detection system** using Linux LKM and Windows KMDF drivers, detecting **SCT/SSDT hooks, DKOM process hiding, LKM concealment, and CR0/LSTAR manipulation** on a live Ubuntu VM.
- Unified kernel–userspace bridge (auto OS-detect) feeds anomaly events through **Ed25519 signing, AES-GCM encryption, Merkle batching, and Solidity/Web3.py blockchain anchoring** for tamper-proof forensic audit trails.
- Tech: **C (Linux LKM / Windows KMDF), Python, SQLite, kprobes, Netlink, Merkle Trees, Web3.py, Solidity**

SecureSplit — Distributed Secure Media Storage [GitHub] Sep 2024 – Oct 2024
Security Engineering • Cryptography • Distributed Storage

- Implemented a **two-share visual cryptography system** — original images are XOR-split into a **local encrypted noise share** and a **key noise share uploaded to IPFS (Pinata)**, ensuring neither share alone reveals any information about the original.
- Tech: **Python, FastAPI, IPFS (Pinata), NumPy, Cryptographic Key Separation**

MatCare — Real-Time IoT Maternal Health Platform [GitHub] Jan 2025 – May 2025
Security Engineering • IoT Systems • Applied ML

- Built a **dual-channel real-time health monitoring backend** ingesting 5 vitals from ESP32 sensors via Blynk APIs for both mother and newborn, with a **Random Forest classifier** for health-state prediction, a **PDF/DOCX clinical report parser** extracting amniotic fluid levels and gestational data, and **Ethereum Sepolia blockchain** registration of patient identities via smart contract.
- Tech: **Python (Flask), PostgreSQL, Blynk APIs, Scikit-learn, Web3.py, Solidity, pdfplumber, ESP32**

EDUCATION

Ramco Institute of Technology Tamil Nadu, India
B.E. Computer Science and Engineering | CGPA: 7.91 Oct 2023 – May 2027

CERTIFICATIONS

- **Cisco Junior Cybersecurity Analyst** – Cisco Networking Academy
- **Certified AppSec Practitioner** (CAP) & **Certified Network Security Practitioner** (CNSP) – The SecOps Group
- **Ubuntu Linux Professional** - Canonical
- **Microsoft Security Essentials (MSEP)** - Microsoft
- **Software Engineer Professional** - HackerRank
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ACHIEVEMENTS

- **IndiaSkills Southern Regionals 2026 (Cyber Security)** — Medallion of Excellence
- **Apoorv CTF 2026** — Rank 44 Worldwide
- **TryHackMe** — Top 5% Global Ranking